

A Counterexample to Conjecture 00000000046

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Claude (Opus 5)[†]

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Abstract

Conjecture 00000000046 asserts that the prime-sum graphs G_n converge, in cut distance, to the constant graphon $W \equiv 1/2$. They do not, and not merely slowly: every G_n is triangle-free, so its triangle density is 0 while that of W is $1/8$, and the counting lemma puts $\delta_\square(W_{G_n}, W) \geq 1/24$ for *every* n . The edge density fails too, tending to 0 rather than $1/2$. The sequence does converge — to the zero graphon.

1 Statement

For $n \geq 1$ let G_n be the graph on $\{1, \dots, n\}$ with $i \sim j$ if and only if $i \neq j$ and $i + j$ is prime. Let W_{G_n} denote the step graphon associated with G_n , let $\delta_\square$ be the cut distance, and for a finite simple graph F let $t(F, \cdot)$ be homomorphism density. Conjecture 00000000046 asserts

$$\delta_\square(W_{G_n}, \tfrac{1}{2}) \longrightarrow 0,$$

where $\frac{1}{2}$ denotes the constant graphon $W \equiv 1/2$.

2 G_n is bipartite, hence triangle-free

Lemma 1. *Let $i \neq j$ be positive integers of the same parity. Then $i \sim j$ fails in every G_n .*

Proof. $i + j$ is even. Being distinct positive integers, $i + j \geq 1 + 3 = 4$ when both are odd and $i + j \geq 2 + 4 = 6$ when both are even; in either case $i + j > 2$. An even number greater than 2 is not prime. $\square$

So every edge of G_n joins an odd vertex to an even one: G_n is bipartite with sides the odd and the even vertices.

Theorem 2. *G_n is triangle-free for every n .*

Proof. Among any three vertices, two have the same parity; by Lemma 1 they are not adjacent. $\square$

3 The conjecture is false, quantitatively

Theorem 3. *For every n ,*

$$\delta_\square(W_{G_n}, \tfrac{1}{2}) \geq \tfrac{1}{24}.$$

In particular $\delta_\square(W_{G_n}, \frac{1}{2}) \not\rightarrow 0$ and Conjecture 00000000046 is false.

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Proof. Since G_n has no loops, every homomorphism $K_3 \rightarrow G_n$ is injective, so $t(K_3, W_{G_n}) = 6 \cdot \#\{\text{triangles of } G_n\}/n^3$, which is 0 by Theorem 2. For the constant graphon, $t(K_3, \frac{1}{2}) = (\frac{1}{2})^3 = \frac{1}{8}$. The counting lemma states

$$|t(F, U) - t(F, U')| \leq e(F) \delta_{\square}(U, U')$$

for graphons U, U' and finite simple F . Taking $F = K_3$, so $e(F) = 3$,

$$\frac{1}{8} = |t(K_3, \frac{1}{2}) - t(K_3, W_{G_n})| \leq 3 \delta_{\square}(W_{G_n}, \frac{1}{2}),$$

which is the claim. $\square$

4 The edge density fails as well

Triangle-freeness already settles the conjecture. The first moment fails too, and in the opposite direction from what $W \equiv 1/2$ would require.

Proposition 4. *Let $d(G_n) = e(G_n)/\binom{n}{2}$. Then*

$$d(G_n) \leq \frac{n+5}{3(n-1)} + \frac{2}{n(n-1)} = \frac{1}{3} + O\left(\frac{1}{n}\right).$$

Proof. Let $i < j \leq n$ with $i+j$ prime. By Lemma 1, $i+j$ is odd. If $3 \mid i+j$ then $i+j$, being prime, equals 3, which happens only for the single pair $\{1, 2\}$. Every other edge therefore has $i+j \equiv 1$ or $5 \pmod{6}$. For fixed i and a fixed residue r , the number of $j \leq n$ with $i+j \equiv r \pmod{6}$ is at most $\lceil n/6 \rceil \leq (n+5)/6$; two residues give at most $(n+5)/3$. Summing over $i \leq n$ counts each edge twice, so $e(G_n) \leq n(n+5)/6 + 1$, and dividing by $\binom{n}{2} = n(n-1)/2$ gives the bound. $\square$

In fact $d(G_n) \rightarrow 0$: each prime $p \leq 2n$ arises as $i+j$ for at most $n/2$ pairs $i < j \leq n$, so $e(G_n) \leq n\pi(2n)/2$ and $d(G_n) \leq n\pi(2n)/(n(n-1)) = O(1/\log n)$ by the prime number theorem. The computed values agree:

n	10	20	50	100	200	400	800	1600
$e(G_n)$	18	56	295	1044	3676	13239	47811	174317
$d(G_n)$	.400	.295	.241	.211	.185	.166	.150	.136

Remark 5 (what the sequence does converge to). Since $W_{G_n} \geq 0$, the cut norm is at most the L^1 norm, so

$$\delta_{\square}(W_{G_n}, 0) \leq \int W_{G_n} = d(G_n) + O(1/n) \rightarrow 0.$$

Thus G_n converges in cut distance to the *zero* graphon $W \equiv 0$, not to $W \equiv 1/2$. A sequence of graphs converges to a constant graphon c only if its edge density tends to c ; here that forces $c = 0$. The conjecture would be correct only for a sequence of density $1/2$, which G_n is not, and the triangle obstruction of Theorem 3 rules out $W \equiv 1/2$ on its own.

5 Formal verification

The accompanying Lean 4 project formalizes the combinatorial core against Mathlib:

- `Adj` — adjacency in the prime-sum graph, $i \neq j$ and $i+j$ prime, with primality Mathlib's `Nat.Prime`.

- `not_adj_of_same_parity`, `parity_ne_of_adj` — Lemma 1: same-parity vertices are never adjacent, so G_n is bipartite between odds and evens.
- `no_triangle` — Theorem 2: no three vertices are pairwise adjacent.
- `triangleCount`, `triangleCount_eq_zero` — the triangle count of G_n , over increasing triples of $\{1, \dots, n\}$, is 0 for every n .

Scope. Mathlib contains no graph-limit theory: there are no graphons, no cut norm or cut distance, and no homomorphism densities. The step from Theorem 2 to Theorem 3 — the counting lemma — is therefore carried out in this paper and is *not* part of the Lean development, which establishes the input to it. We state this explicitly rather than leave the coverage of the formalization to be inferred. Building the missing theory is a substantial project in its own right and is not attempted here.

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with Mathlib `v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms triangleCount_eq_zero` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.